

Nama : Ricy Rifani Putra

Nim : A11.2025.16464

Materi : Stack and Queue

Output

```
▼ TERMINAL
• PS C:\MATKUL (TI) UDINUS SEM-2\ASD\Cpp> g++ stack-11/stack/main.cpp -o main
• PS C:\MATKUL (TI) UDINUS SEM-2\ASD\Cpp> ./stack-11/stack/main.exe
stack / tumpukan buku
[harry pooter|jk|0x1852428d970] top
    |
    V
[laskar|andrea|0x1852428d5b0]
    |
    V
[sporsi mie ayam sblum mati|brian|0]

Queue dengan class
[Sasuke | 0x18524293840] --> [Sakura | 0]
[Naruto | 0x18524293a50] --> [Sasuke | 0x18524293840] --> [Sakura | 0]
Calon Hokage urutan pertama adalah Naruto
Maaf, kapasitas penuh
[Kakashi | 0x18524293ab0] --> [Naruto | 0x18524293a50] --> [Sasuke | 0]
Calon Hokage urutan pertama adalah Kakashi
❖ PS C:\MATKUL (TI) UDINUS SEM-2\ASD\Cpp> 
```

Source Code

```
stack-II > stack n queue > C++ main.cpp
1  // Nama : Ricy Rifani Putra
2  // Nim  : A11.2025.16464
3
4  #include <iostream>
5  using namespace std;
6
7  class Nodestack{
8  public:
9      string judul,penulis;
10     Nodestack *next;
11
12     Nodestack(){
13         judul   = "";
14         penulis  = "";
15         next    = NULL;
16     }
17
18     Nodestack(string judul,string penulis){
19         this -> judul = judul;
20         this -> penulis = penulis;
21         next = NULL;
22     }
23 };
24 class stack{
25 private:
26     Nodestack *top;
27
28 public:
29     stack(){
30         top = NULL;
31     }
32     bool isEmpty();
33     void push(string judul, string penulis);
34     //tumpuk atau tambah
35     void pop(); //ambil atau hapus
36     void cetak();
37 };
38
39 class Node {
40 public:
41     string nama;
42     Node *next;
43
44     Node() {
45         nama = "";
46         next = NULL;
47     }
48 }
```

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46     Node(string nama) {
47         this->nama = nama;
48         this->next = NULL;
49     }
50 };
51
52
53 class Queue {
54 private:
55     Node *head;
56     Node *tail;
57     int kapasitas;
58
59 public:
60     Queue() {
61         head = NULL;
62         tail = NULL;
63         kapasitas = 100;
64     }
65
66     Queue(int kapasitas) {
67         head = NULL;
68         tail = NULL;
69         this->kapasitas = kapasitas;
70     }
71
72     bool isEmpty() {
73         return head == NULL;
74     }
75
76     bool isFull() {
77         return panjangQ() >= kapasitas;
78     }
79
80     int panjangQ() {
81         int count = 0;
82         Node *temp = head;
83         while (temp != NULL) {
84             count++;
85             temp = temp->next;
86         }
87         return count;
88     }

```

```

91     void enqueue(string nama) {
92         if (isFull()) {
93             cout << "Maaf, kapasitas penuh" << endl;
94             return;
95         }
96         Node *newNode = new Node(nama);
97         if (isEmpty()) {
98             head = newNode;
99             tail = newNode;
100         } else {
101             tail->next = newNode;
102             tail = newNode;
103         }
104     }
105
106
107     void enqueuePriority(string nama) {
108         if (isFull()) {
109             dequeueBelakang(); // Sakura tergeser keluar
110         }
111         Node *newNode = new Node(nama);
112         newNode->next = head;
113         head = newNode;
114     }
115
116     // Hapus depan
117     void dequeue() {
118         if (isEmpty()) {
119             cout << "Queue kosong!" << endl;
120             return;
121         }
122         Node *temp = head;
123         head = head->next;
124         if (head == NULL) tail = NULL;
125         delete temp;
126     }
127
128     // Hapus belakang
129     void dequeueBelakang() {
130         if (isEmpty()) {
131             cout << "Queue kosong!" << endl;
132             return;
133         }
134         if (head == tail) {

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134         if (head == tail) {
135             delete head;
136             head = NULL;
137             tail = NULL;
138             return;
139         }
140         Node *temp = head;
141         while (temp->next != tail) {
142             temp = temp->next;
143         }
144         delete tail;
145         tail = temp;
146         tail->next = NULL;
147     }
148
149     // Lihat elemen terdepan
150     string peek() {
151         if (isEmpty()) return "";
152         return head->nama;
153     }
154
155     // Cetak semua isi queue
156     void cetak() {
157         if (isEmpty()) {
158             cout << "Queue kosong!" << endl;
159             return;
160         }
161         Node *temp = head;
162         while (temp != NULL) {
163             cout << "[" << temp->nama << " | " << temp->next << "]";
164             if (temp->next != NULL) cout << " --> ";
165             temp = temp->next;
166         }
167         cout << endl;
168     }
169 };
170
171 int main(){
172
173     cout << " stack / tumpukan buku\n";
174     stack s;

```

```

176     s.push ("sporsi mie ayam sblum mati", "brian");
177     s.push ("laskar", "andrea");
178     s.push ("harry pooter", "jk");
179
180     //s.pop();
181     s.cetak();
182     cout << endl;
183     cout << endl;
184     cout << endl;
185
186
187     // QUEUE
188     cout << "Queue dengan class" << endl;
189     Queue hokage(3);
190
191     hokage.enqueue("Sasuke");
192     hokage.enqueue("Sakura");
193     hokage.cetak();
194
195     hokage.enqueuePriority("Naruto");
196     hokage.cetak();
197     cout << "Calon Hokage urutan pertama adalah " << hokage.peek() << endl;
198
199     hokage.enqueue("Boruto");
200     hokage.enqueuePriority("Kakashi");
201     hokage.cetak();
202     cout << "Calon Hokage urutan pertama adalah " << hokage.peek() << endl;
203
204     return 0;
205 }
206
207 bool stack::isEmpty(){
208     return top == NULL;
209 }
210
211 void stack::push(string judul,string penulis){
212     Nodestack *temp = new Nodestack (judul,penulis);
213     if (isEmpty()){
214         top=temp;
215     }else{
216         temp->next=top;
217         top=temp;
218     }
219 }

```

```

220 |
221 | void stack::pop(){
222 |     if(!isEmpty()){
223 |         Nodestack *temp = top;
224 |         top = top->next;
225 |         delete temp;
226 |     }
227 | }
228 |
229 | void stack::cetak(){
230 |     Nodestack *temp = top;
231 |     while(temp != NULL){
232 |         cout << "[" << temp->judul << "|" << temp->penulis << "|" << temp->next << "]";
233 |
234 |         if(temp == top && temp->next != NULL){
235 |             cout << "\t top\n\t | \n\t \\/\n";
236 |         } else if (temp == top) {
237 |             cout << "\n\t top\n\t | \n\t \\/\n";
238 |         } else if (temp->next != NULL){
239 |             cout << "\n\t | \n\t \\/\n";
240 |         }
241 |         temp = temp->next;
242 |     }
243 | }

```

**** Terimakasih ****